Experiment 7: Mean-Shift Based Image Segmentation

Objective: To segment images using the Mean-Shift algorithm based on both color and spatial features.

Algorithm (Image-based Mean-Shift):

1. Load the input image and convert it from BGR to RGB.
2. Prepare the feature space using R, G, B color channels and pixel coordinates X and Y.
3. Estimate the bandwidth for Mean-Shift clustering using a sample of the feature space.
4. Apply the Mean-Shift algorithm on the full feature space.
5. Assign each pixel to its corresponding cluster centroid to reconstruct the segmented image.
6. Display both the original and segmented image.

Algorithm (Video-based Mean-Shift):

1. Open the video file and set up video writer to save the output.
2. Read the first frame and estimate Mean-Shift bandwidth using a downscaled version.
3. For each frame in the video:

a. Resize the frame and extract color features.
b. Apply Mean-Shift clustering using the estimated bandwidth.
c. Reconstruct the segmented image from the cluster centers.
d. Resize the segmented frame to original size and overlay cluster info.
e. Show both original and segmented frame; save the segmented frame.

4. Continue until video ends or user interrupts.
5. Release video resources and close all windows.

Theory:
Mean-Shift is a clustering technique based on kernel density estimation. Unlike KMeans, it does not require the number of clusters to be specified beforehand. It works by shifting a window towards the region of the highest data point density (modes of the data).

In image segmentation, combining spatial information (X, Y coordinates) with color information (RGB) allows the Mean-Shift algorithm to create coherent regions that respect object boundaries and spatial continuity.

Flowchart:

Start → Load Image/Video → Extract Color & Position Features → Estimate Bandwidth → Apply Mean-Shift Clustering → Form Segmented Output → Display & Save → End